Trade Data Quality Diagnostic Report

Brazil ↔ Canada – Imports & Exports

Period: Jan 2024 – Sep 2025

Generated: 2026-02-17 03:20

Executive Summary

IMPORT DATASET: 63,716 valid records

- Zero Quantity rows: 27,481 (43.1%)
- Outliers detected: 4,235
- Products with multiple UOMs: 0

EXPORT DATASET: 14,964 valid records

- Zero Quantity rows: 9,498 (63.5%)
- Outliers detected: 788
- Products with multiple UOMs: 2

MAIN DIAGNOSTIC HIGHLIGHTS:

- Zero-quantity records dominate data quality issues
- Unit price calculation blocked for large % of rows
- Very wide price ranges → category-specific analysis essential
- Inconsistent units of measure on some products

PDF Roadmap – Sections & Content

Page 1 **Title Page**

Page 2 **Executive Summary**

Page 3 **Document Roadmap (this page)**

Import Data Diagnostic (pages ~4–11):

- Missingness Analysis
- Zero Quantity Dashboard
- Unit Price Distribution
- Price by HS Chapter (Boxplot)
- Outlier Analysis
- Unit of Measure Consistency
- Temporal Trends (Zero Qty %)
- Non-ASCII / Special Characters

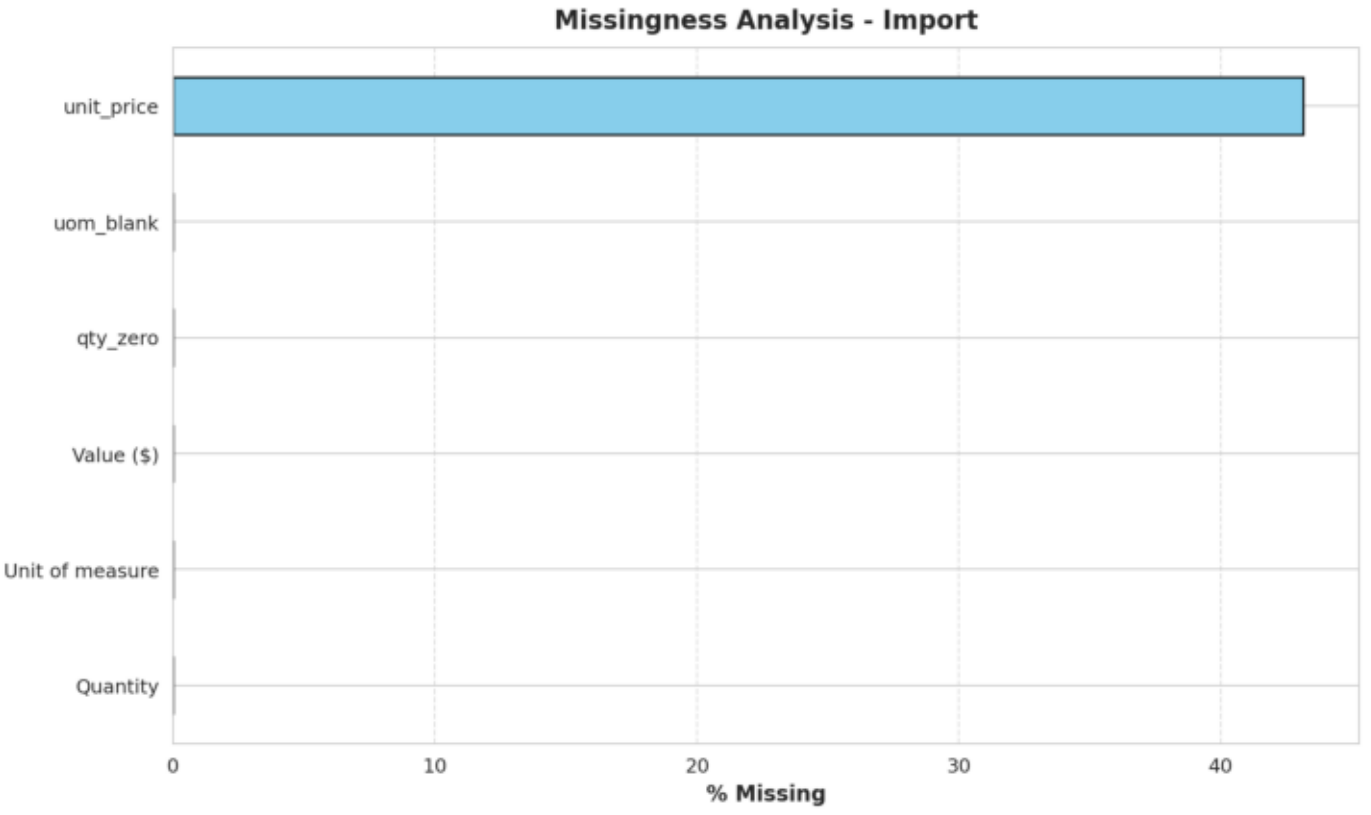
Export Data Diagnostic (pages ~12–19):

Same 8 views as above, but for Export data

Diagnostic focus: identify issues + recommended actions per finding

Each chart page includes tailored next-step recommendations

Missingness Analysis – Import



Key Findings:

- Unit price missing in ~43.1% of records
- Almost entirely caused by Quantity = 0
- Core identifier fields (Period, Commodity, Country) nearly complete

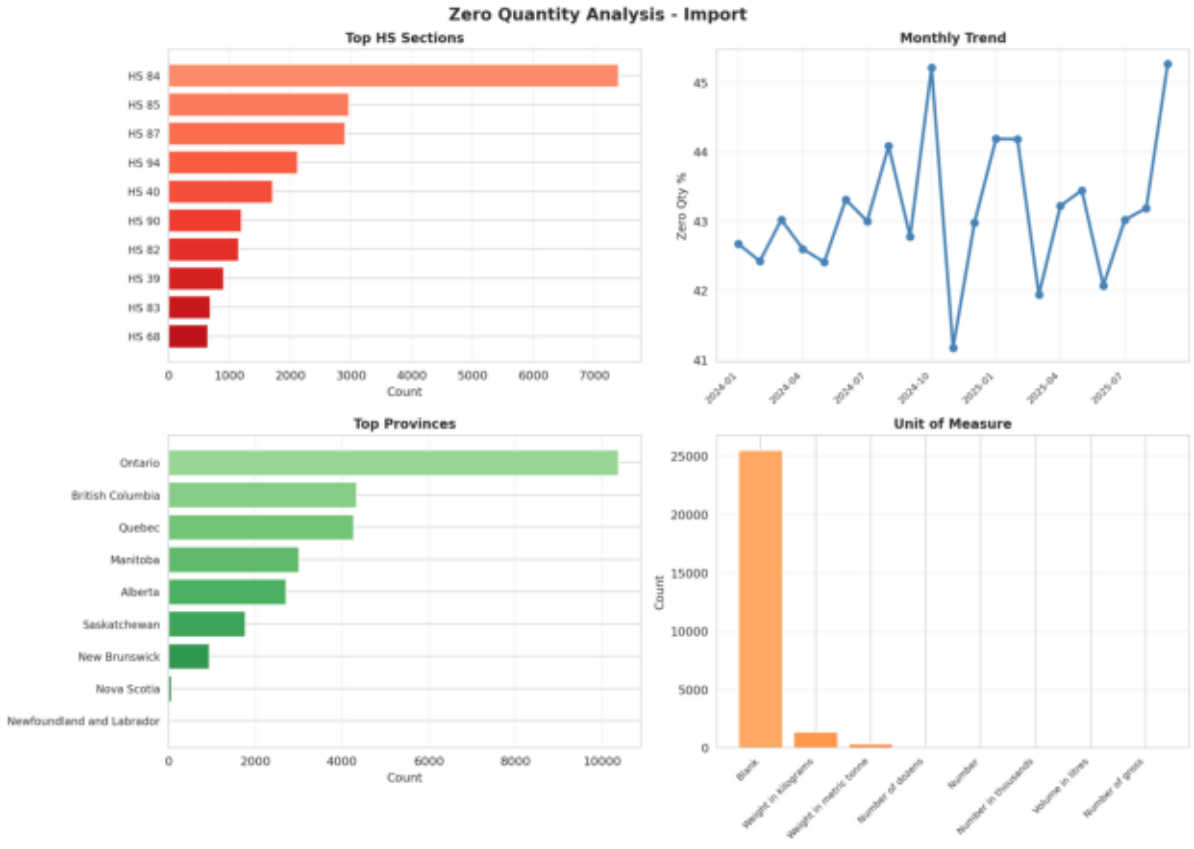
Recommended Next Steps:

→ Check source: are zero-quantity rows intentional (e.g. estimates, samples)?

→ Flag or exclude these rows from unit price calculations

→ Consider separate reporting layer for zero-volume trades

Zero Quantity Analysis – Import



Key Findings:

- 43.1% of records show Quantity = 0
- Pattern consistent across months – not seasonal
- Higher concentration in specific HS sections

Recommended Next Steps:

→ Sample 50–100 zero-quantity records to understand business meaning

→ Decide: impute, flag, exclude, or report separately?

→ Prioritize high-value zero rows for manual review

Unit Price Distribution – Import

Key Findings:

- Extremely wide range – log scale required
- Most prices fall between 10–1,000
- Long tail of very low/high unit prices

Recommended Next Steps:

→ Review records with unit price $< 0.01 or > 10{,}000$ for validity

→ Consider HS-section specific price thresholds

→ Flag suspicious extremes before aggregation

Price by HS Chapter – Import

Key Findings:

- Distinct price levels across HS chapters
- High within-chapter variation
- Machinery/electronics often show higher medians

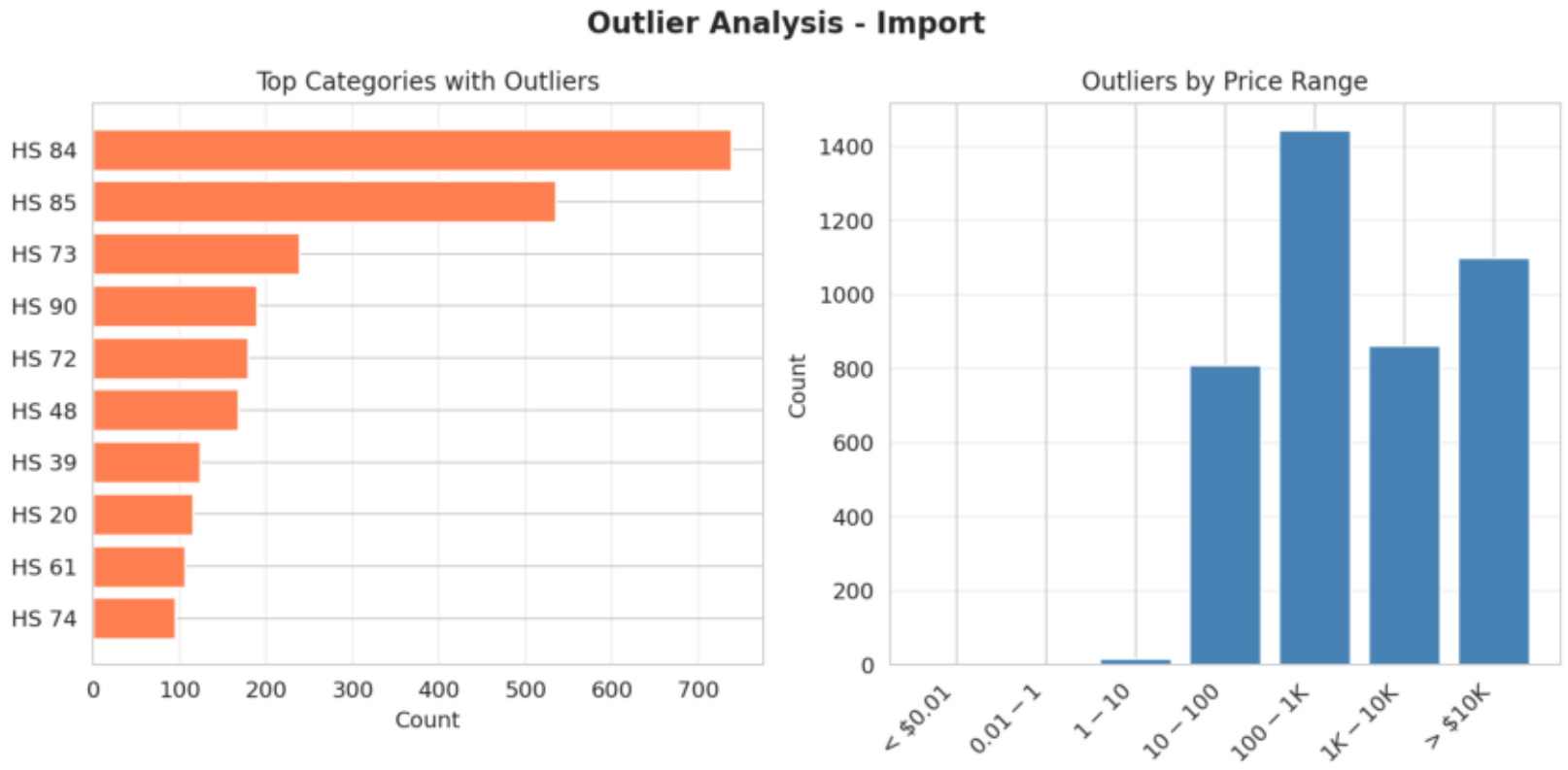
Recommended Next Steps:

→ Apply outlier detection **within** each HS section (not globally)

→ Validate extreme values per chapter with domain knowledge

→ Use chapter-specific medians for imputation if needed

Outlier Analysis – Import



Key Findings:

- 4,235 category-specific outliers detected
- Mostly high-price extremes
- Spread across multiple HS sections

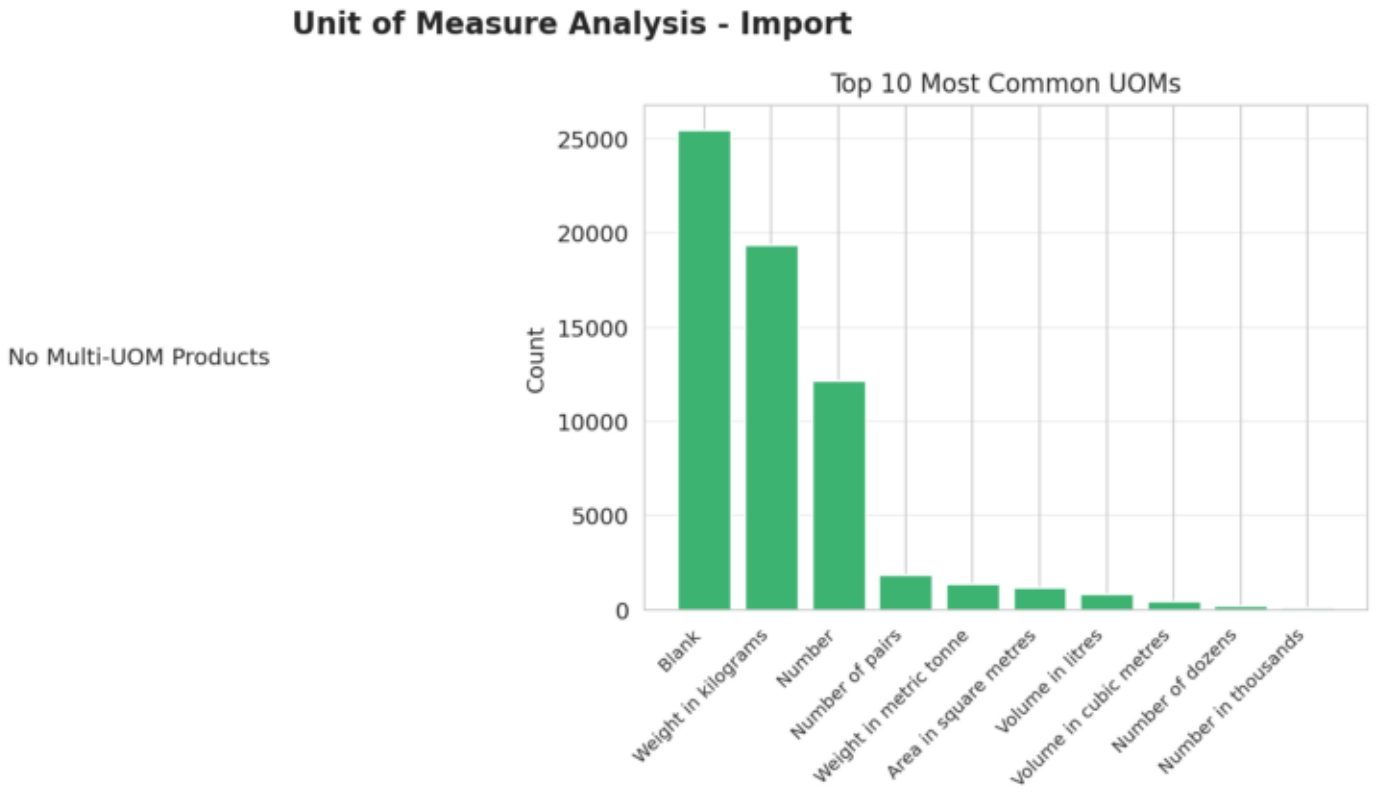
Recommended Next Steps:

→ Export top 100 outliers (by value or deviation) for manual review

→ Classify: real high-value items or data errors?

→ Consider winsorization or filtering only after validation

Unit of Measure Analysis – Import



Key Findings:

- 0 products appear with multiple units
- Prevents accurate quantity summation
- Kg, units, liters, tons most common conflicts

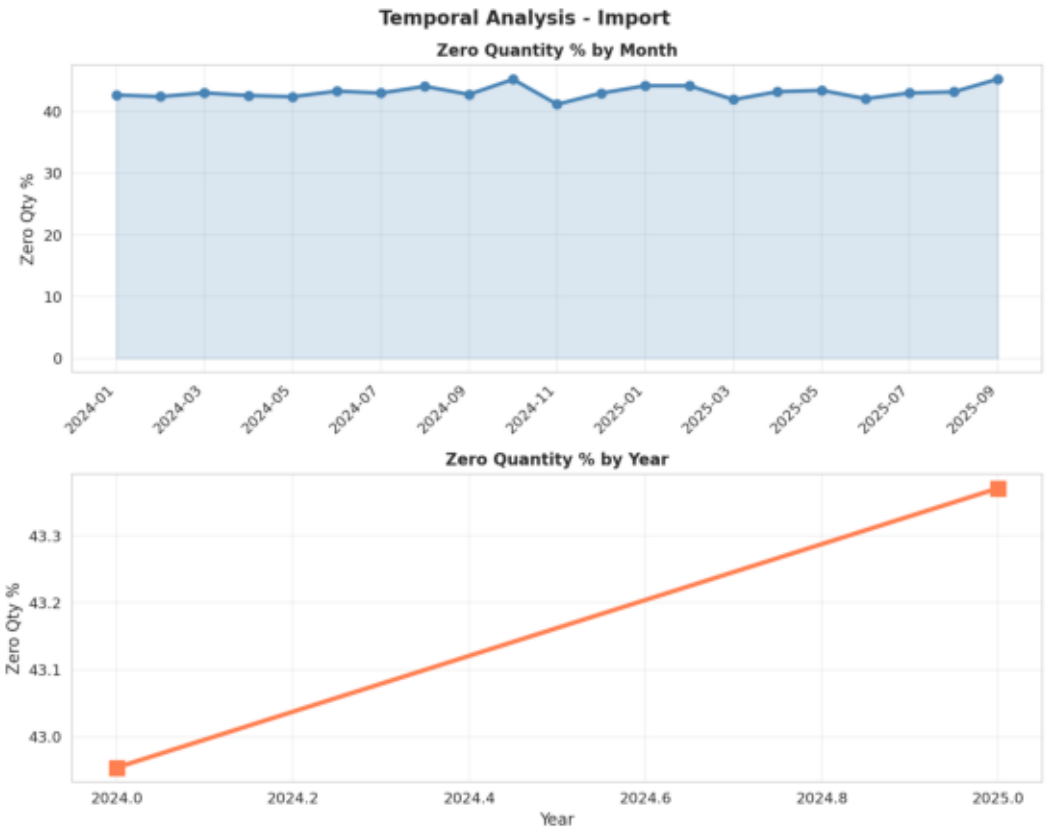
Recommended Next Steps:

→ Prioritize top 20 multi-UOM products by total value for review

→ Verify with source system: intentional or error?

→ Build limited conversion table for high-impact commodities

Temporal Analysis – Import



Key Findings:

- Zero-quantity rate relatively stable over time
- No major anomalies or spikes visible
- Mild seasonal variation may exist

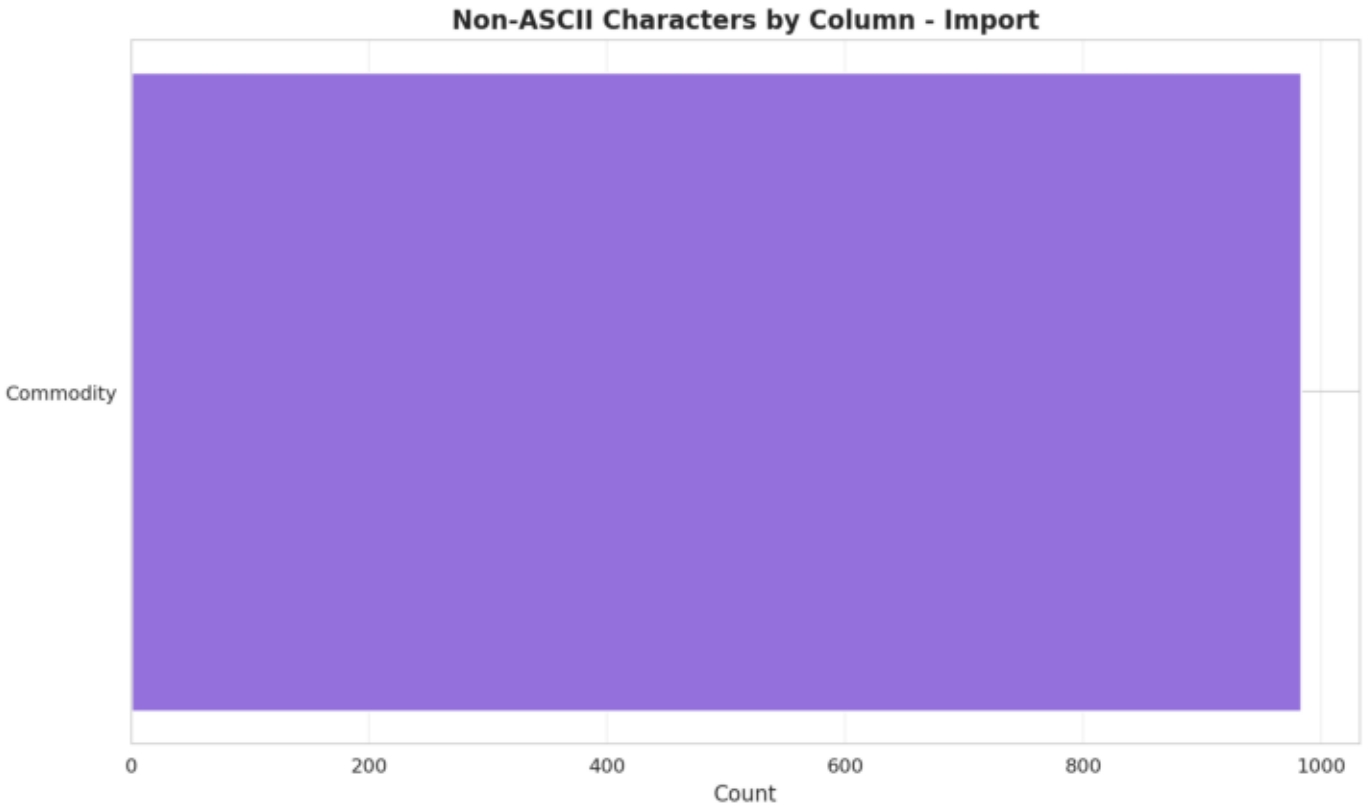
Recommended Next Steps:

→ Confirm temporal coverage is complete for all periods

→ Add month/quarter grouping for future trend reports

→ Monitor for any emerging patterns in next data refresh

Non-ASCII Characters – Import



Key Findings:

- 983 records contain non-ASCII characters
- Mainly in commodity descriptions and location names
- Low overall prevalence

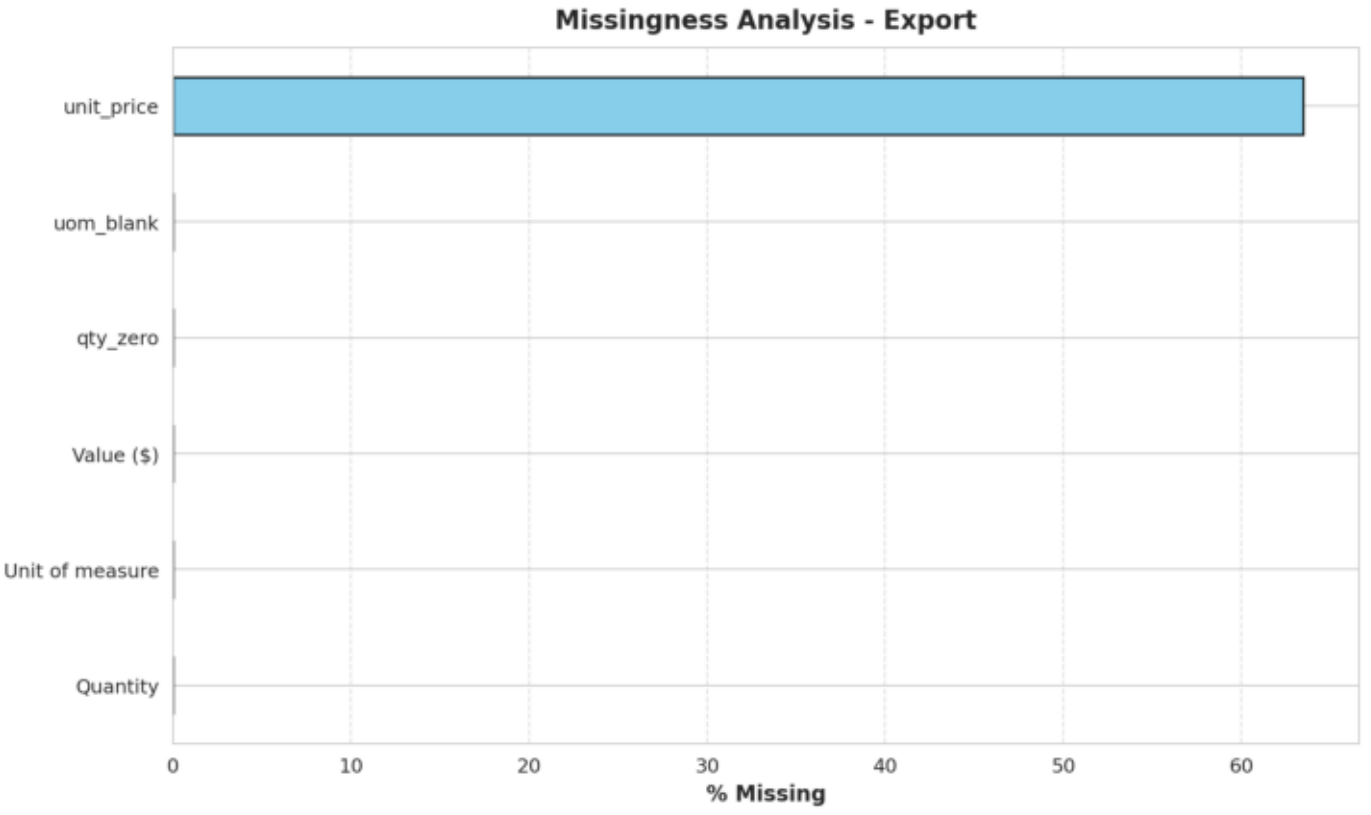
Recommended Next Steps:

→ Test export / matching processes with affected records

→ Apply unidecode normalization only if issues occur

→ Low priority unless downstream systems fail

Missingness Analysis – Export



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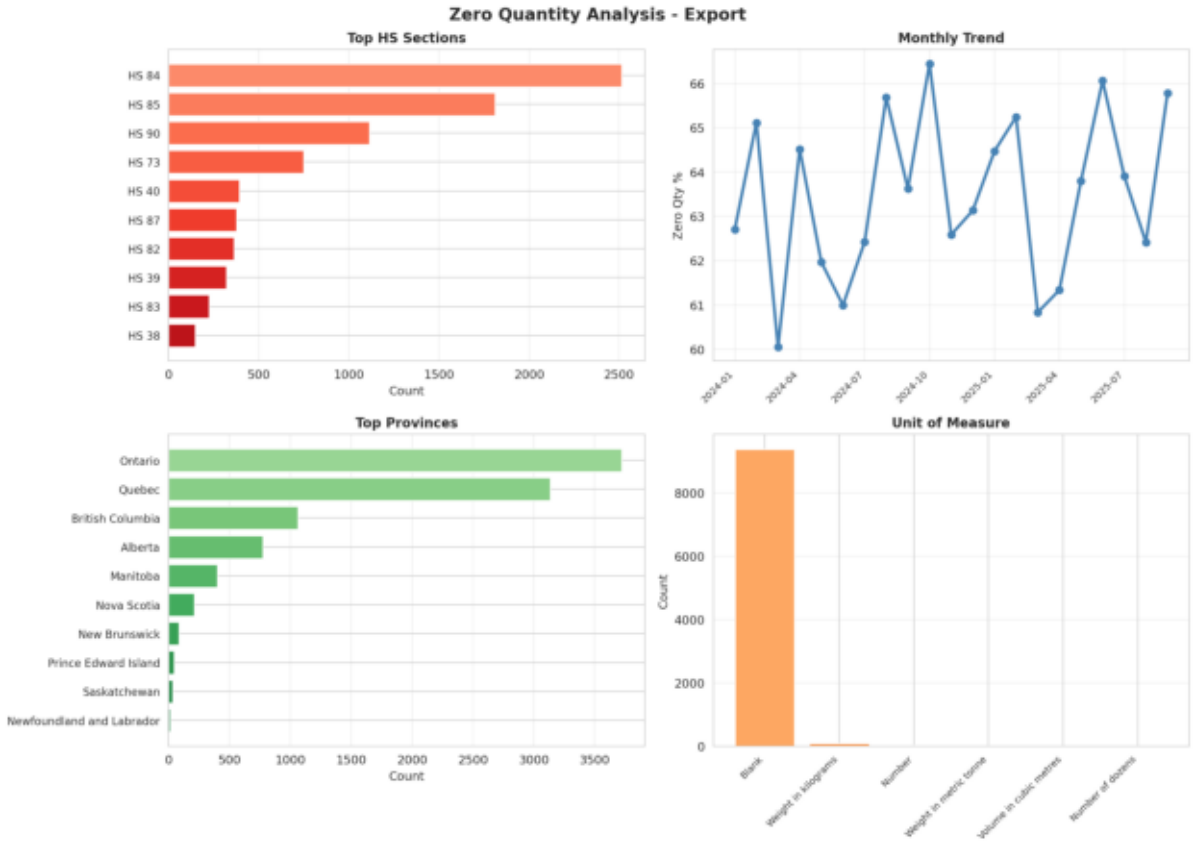

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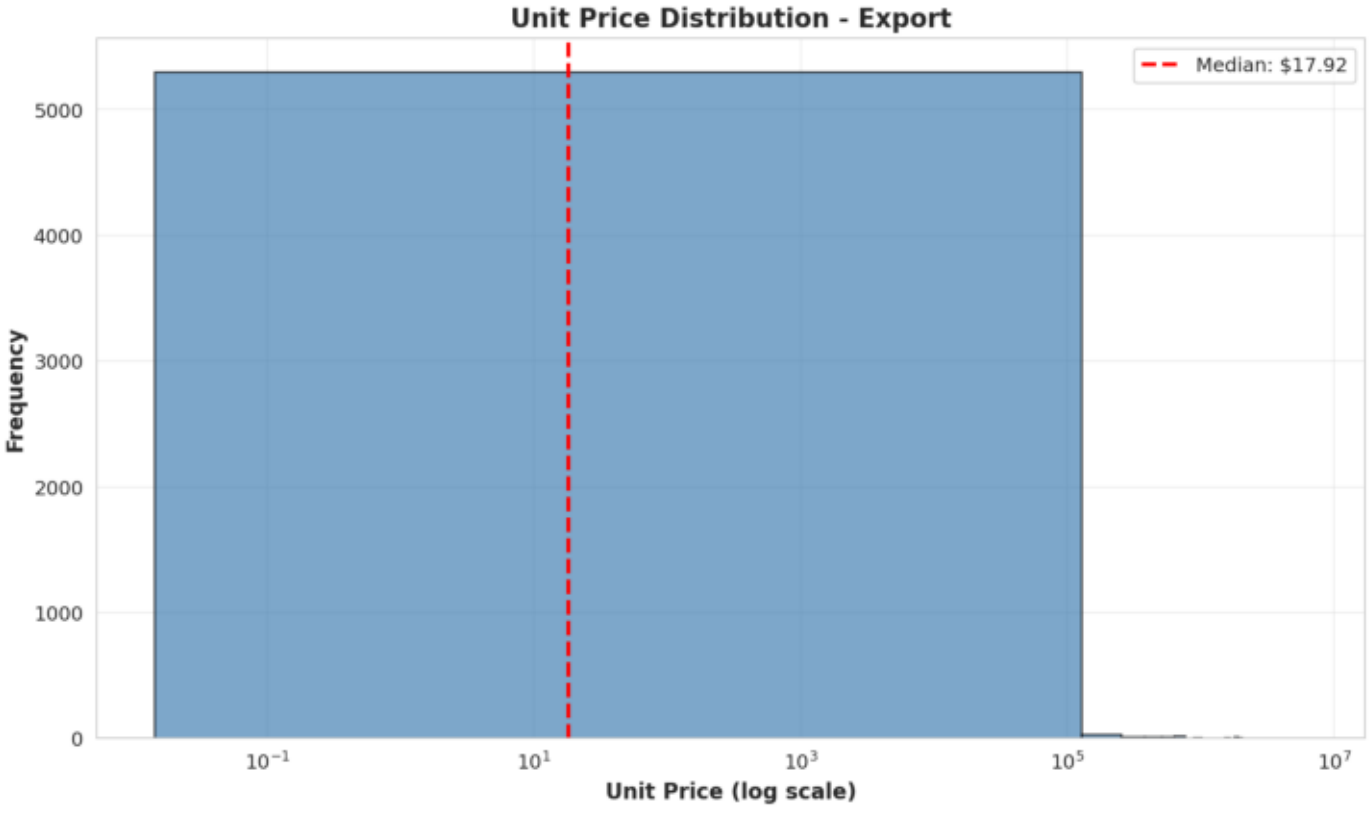

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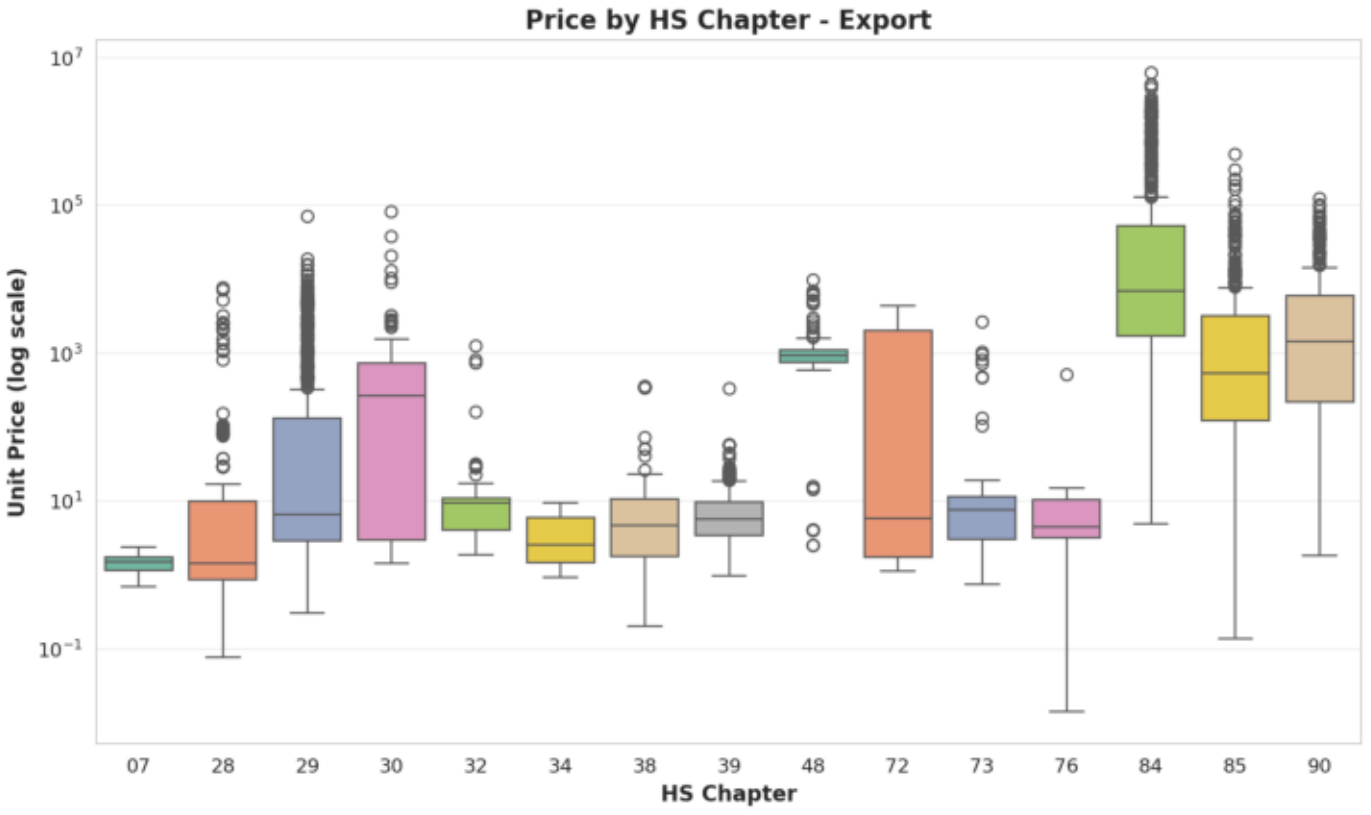
Price by HS Chapter – Export



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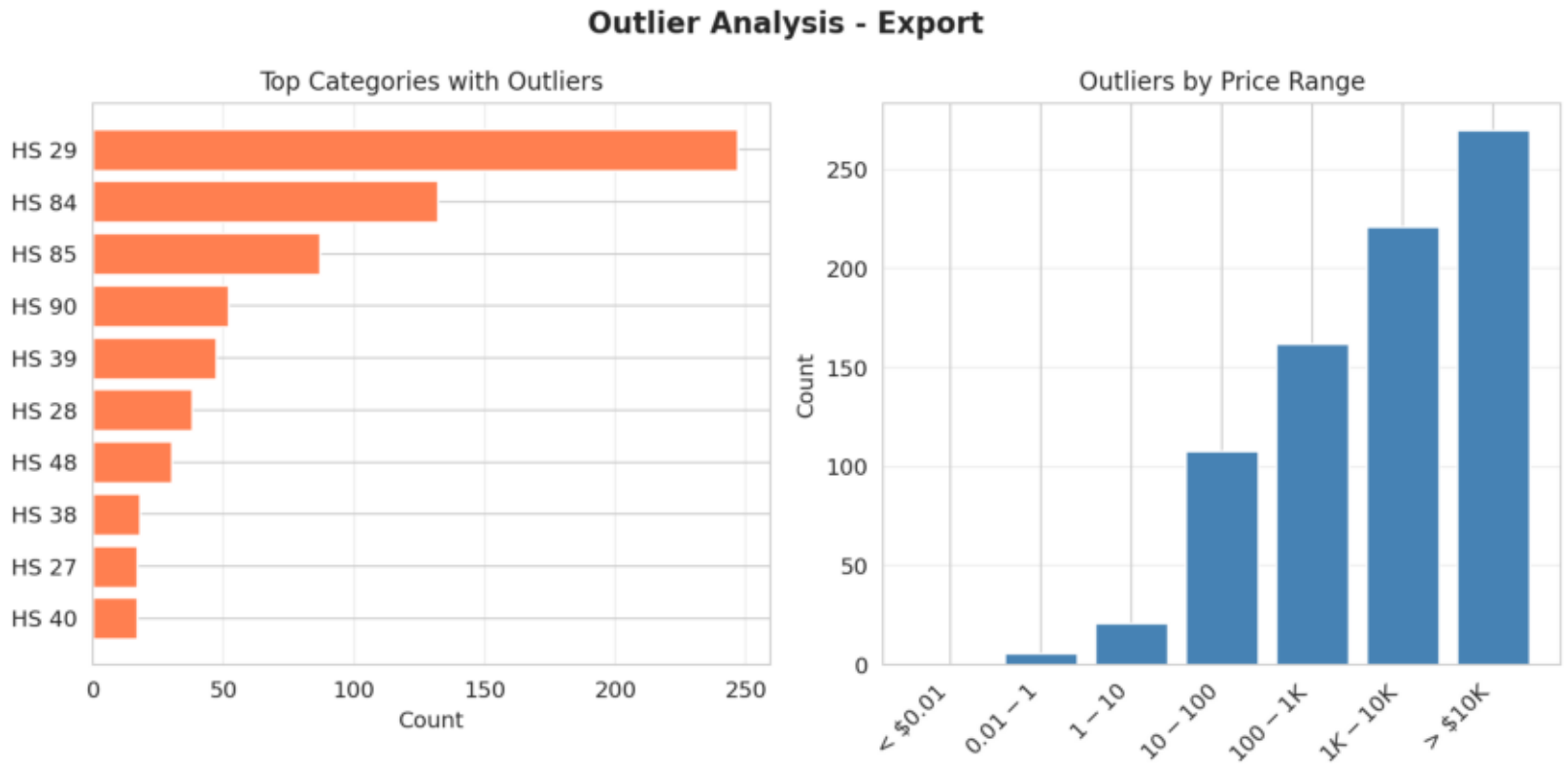

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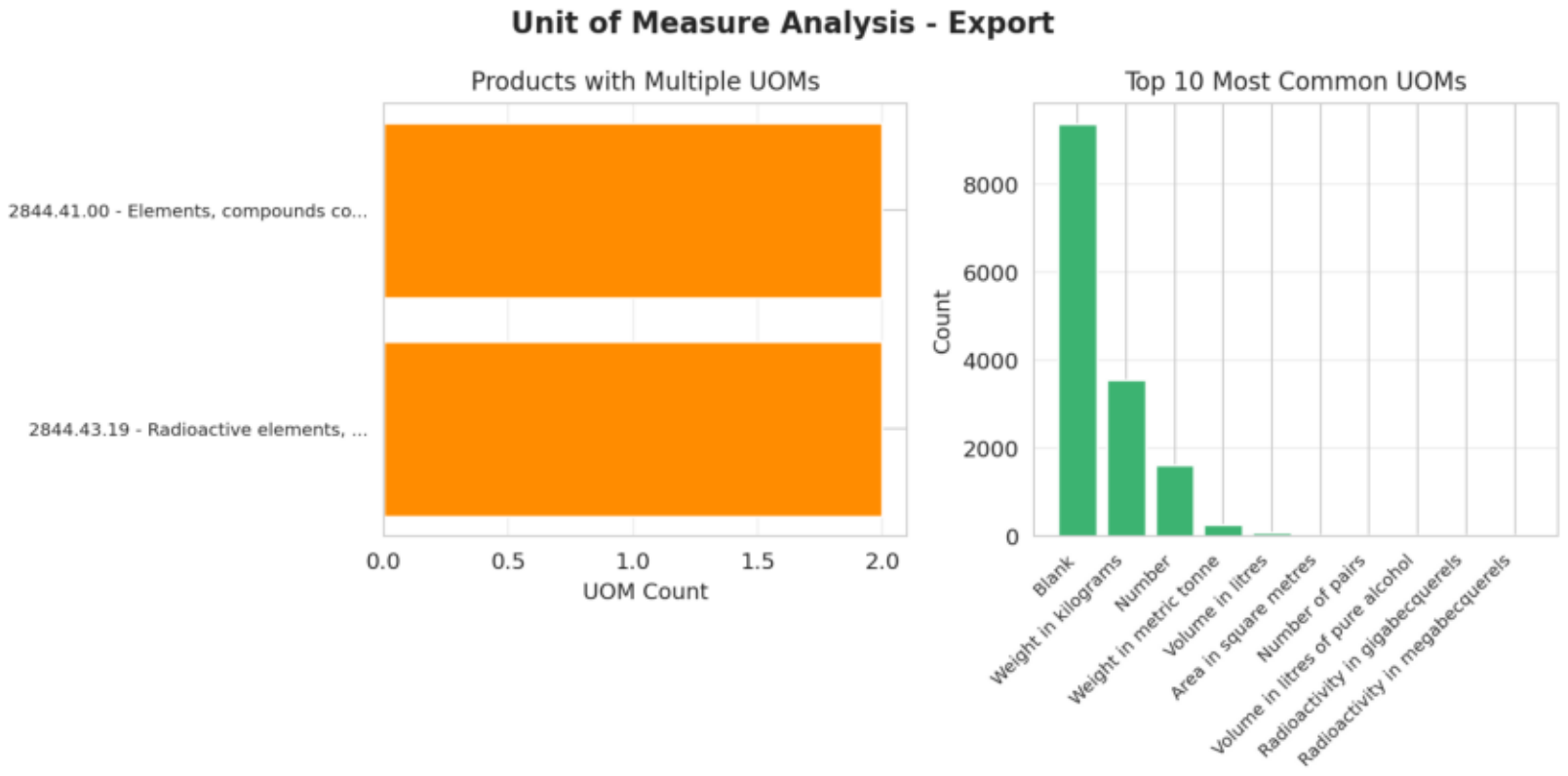

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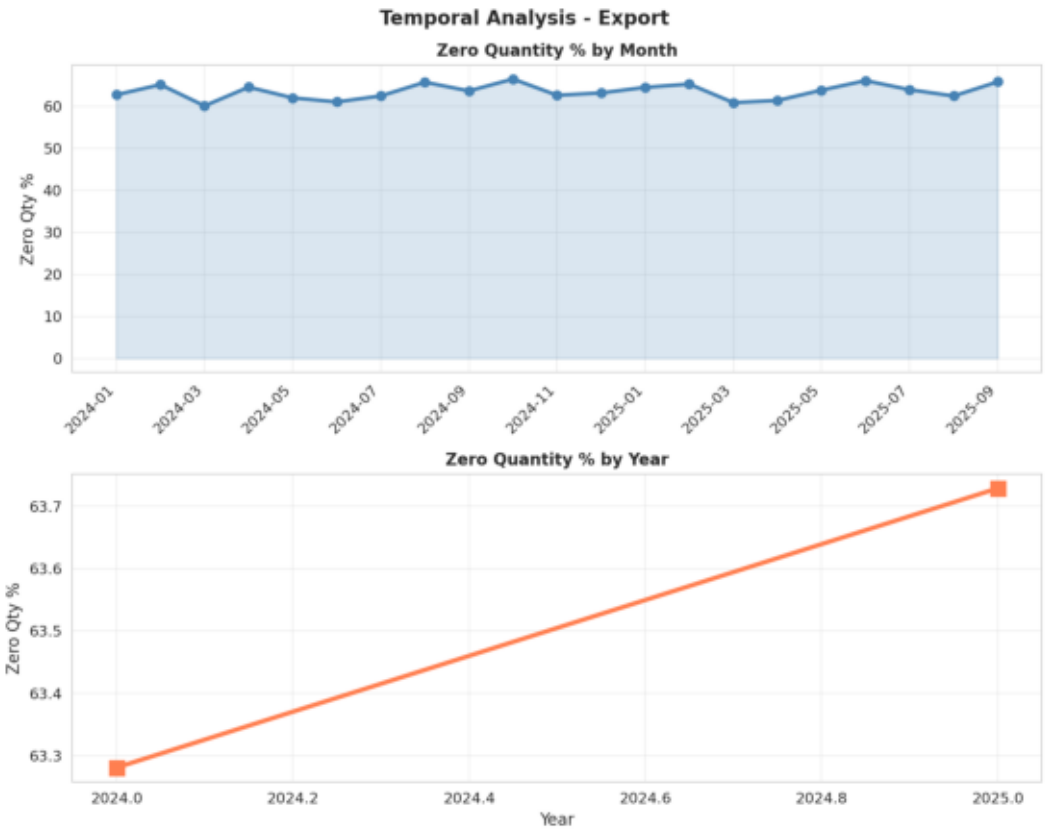

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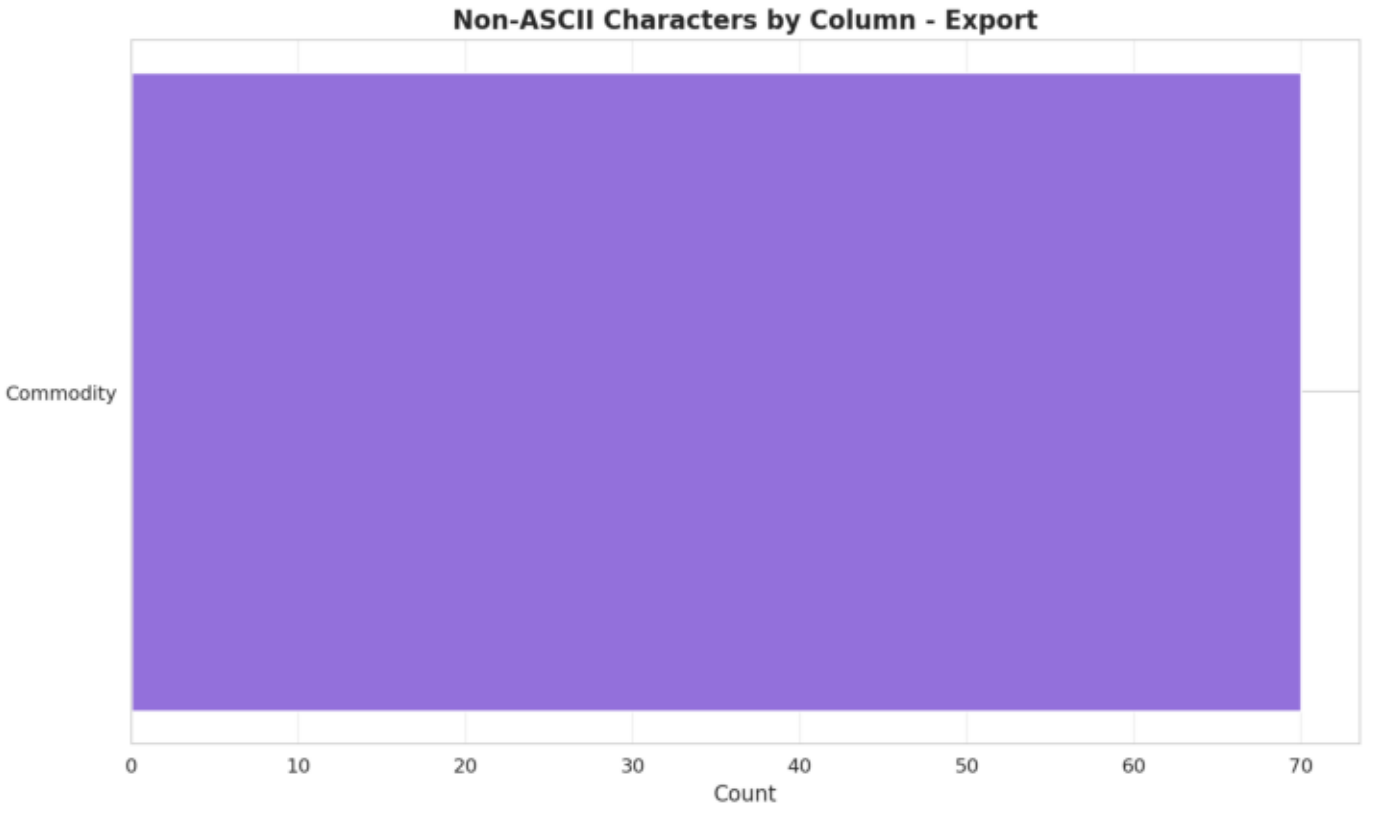

Key Findings:

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- Mainly in commodity descriptions and location names
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